

PaperPing: A Socially-aware AI Agent that Recommends Academic Papers to Research Group Chats with Contextualized Explanations

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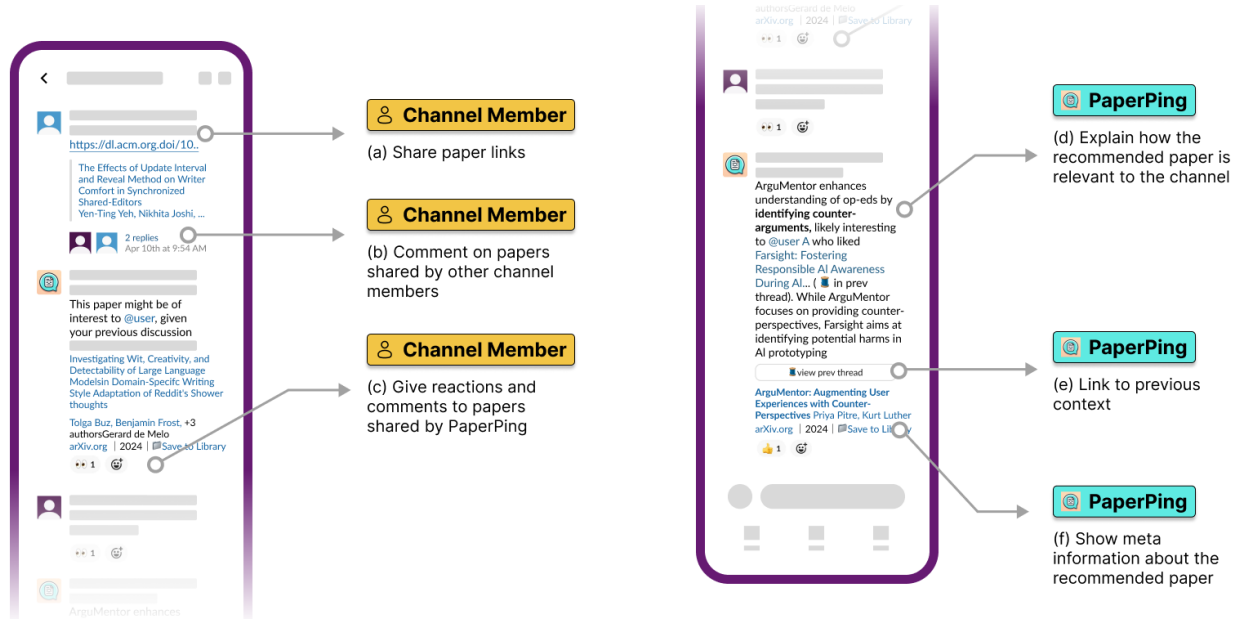


Figure 1: A demonstration of how users interact with PAPERPING in Slack. Channel members share paper links, react to papers shared by other members or PAPERPING and comment on shared papers. After gathering and processing this information, PAPERPING sends new contextually grounded paper recommendations, with explanations of how the recommended papers are relevant to the channel. It also provides links to previous related discussions and includes meta-information about the recommended papers in the recommendation message.

Abstract

AI agents are entering group spaces to facilitate or contribute to conversations. However, without an understanding of group context, these contributions can be irrelevant or even intrusive. In this demo, we introduce a socially-aware AI agent that proactively contributes to research group chats by posting context-aware messages

that contain paper recommendations and contextualized explanations. To achieve this, PaperPing instantiates Social-RAG [14]. It retrieves relevant interactions in the group context, builds a social knowledge base, and extracts social signals to contextualize the generated messages. PaperPing has been deployed in 18 channels as part of a user study for at least 3 months, reaching 500+ researchers. As part of the demo, we will demonstrate PaperPing in a Slack workspace open to all CSCW attendees during and after the conference, allowing researchers to explore the interactions in real settings with the side benefits of fostering collaborative learning within the community.



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CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → **Collaborative and social computing**.

Keywords

AI agent, group communication, retrieval augmented generation, recommender systems, large language models

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1 Introduction

Increasingly, AI-enabled agents are present in online social environments, where they serve as assistants, facilitators, or collaborators in group conversations and contribute to the conversation to support collaborative learning [2], brainstorming [13], or decision-making [16]. A significant risk of AI's direct participation in group spaces is that they can easily be perceived as irrelevant [15] or intrusive [12] to group members, who use the same space to communicate with each other [1, 2]. And when they inevitably fail, existing systems often do not provide easy ways for group members to give feedback in the moment [1], eventually leading users to ignore the messages or even abandon the system [1, 13].

To explore socially-aware AI participation in collaborative spaces, we built an AI agent, **PAPERPING**, which can contribute to scholarly conversations in Slack by proactively posting scholarly paper recommendations and LLM-generated explanations into group chat spaces. Scientific research is collaborative in nature [5, 9], where exchanging and discussing scholarly articles is a common practice and important for conducting high-quality research [5]. Moreover, there has been a rapid growth of academic publications in recent years, which creates significant challenges for researchers to stay on top of relevant papers. **PAPERPING** attempts to address this challenge by surfacing relevant papers during research conversations in a socially aware way. Instead of following rigid message templates and pre-set rules which limit AI agents' ability to adapt their content to align with evolving group preferences and specific social contexts [4, 8, 11], **PAPERPING** adopts a flexible and contextualized approach, **Social-RAG**, which allows LLM-based AI agents that takes social signals (e.g., topical preferences) gleaned from prior interactions between group members and between members and the AI agent to contextualize and adapt the agent's generation. Much like how traditional RAG pipelines use factual knowledge bases to improve factual grounding [6, 7, 10], Social-RAG retrieves and processes "social facts" from a social knowledge base to improve an agent's social grounding.

PAPERPING has been successfully deployed in 18 research Slack channels, reaching over 500 researchers as part of a user study [14], where we find that PAPERPING effectively learns researchers' individual and group preferences with minimum upfront effort from

users, and its generated messages are considered contextually relevant. Moreover, the contextual awareness enhanced the perceived value of PAPERPING's contribution in the group and helped foster common ground and user engagement in some groups.

In this demo, we hope to invite a broader audience from CSCW to experience and use PAPERPING. Specifically, we will create a dedicated Slack workspace that's open to all CSCW attendees and researchers to share and discuss research, with the facilitation of PAPERPING. We will start advertising the workspace and building the community before the conference. Participants will be invited to initiate and participate in topic-based channels (e.g., Human AI interaction, future of work, etc.) to discuss and share relevant research papers. PaperPing will join each channel to provide proactive and contextualized paper recommendations. At the demo session, participants will be able to browse and even join these discussions, interact with PaperPing by reacting and commenting on its recommendations, and experience how PaperPing could adapt to the different group contexts. This demo will not only showcase the features and interactions of PaperPing in an authentic context, inspire discussion around AI participation in collaborative spaces but will also be a valuable community-building effort for the CSCW community.

2 System overview

2.1 PAPERPING Process and Features

PAPERPING is a Slack App. To use PAPERPING, a channel member will need to invite it to the channel, which gives it access to the channel's interaction history. Upon joining, PAPERPING will share an introductory message to inform channel members about its functions and access to the channels' conversation. Channel members will be able to set a frequency for PAPERPING's recommendation as they prefer (e.g., once a week, once every other day, or every day). Any channel member will be able to remove PAPERPING from their channels at any time, if they wish.

2.1.1 Step 1: Collect and Index interactions to create a social knowledge base. Upon joining the channel, PAPERPING will build a social knowledge base by parsing chat history to extract previously shared papers, user reactions, comments, and interaction patterns. PAPERPING then augments it with the Semantic Scholar database which provides additional relevant metadata (e.g., authors, venue, paper citation relationship). PAPERPING also optionally allows users to link their group chat account to their academic publication record, letting it know the user's affiliations and publication history. When PAPERPING prepares to make a recommendation, it sends recent items from this paper database that have received engagement in the channel (e.g., received emoji reactions or comments, or shared by human users) to an off-the-shelf paper recommender system to retrieve the next recommended paper.

2.1.2 Step 2: Retrieve social signals. After accessing a paper, PAPERPING next retrieves social signals related to it to contextualize the message. It focuses on three groups of social signals: (1) Connections from the new paper to previously shared papers in the chat channel, looking for potential topic overlaps, citation relationships, or common authorships. (2) Metadata items about the paper, such

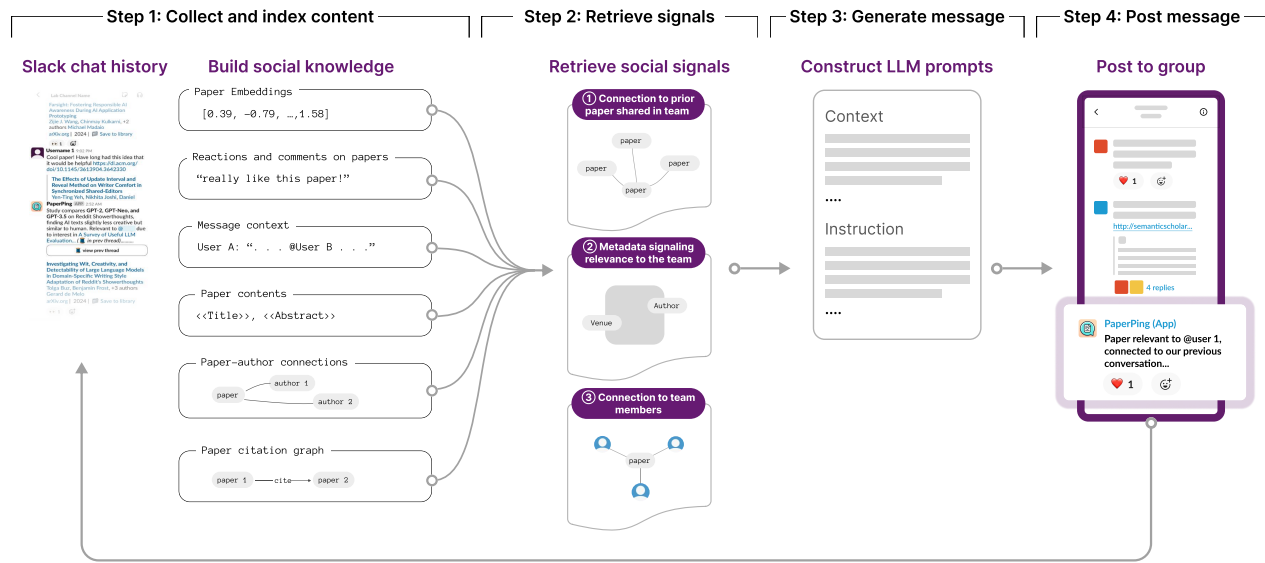


Figure 2: The PAPERPING implementation, leveraging the Social RAG workflow. PAPERPING extracts relevant social information from two data sources: the Semantic Scholar knowledge base, providing paper content, paper-author connections, and citation graphs; and Slack chat history, which includes previously mentioned papers, emoji reactions, comments, and user interactions (Step 1). PAPERPING then retrieves relevant social signals from the data using three heuristics (relevant metadata, the most similar and discussed prior paper in the chat, and the member most likely to be interested) developed over a three-month iterative design process (Step 2). Next, PAPERPING uses chains of LLM prompts to turn the retrieved social signals into a natural language explanation (Step 3). Finally, PAPERPING posts the message to research group channels, where group members can provide emoji reactions or reply to the message to give feedback (Step 4).

as authors or publisher, that have some connection to prior interactions in the chat channel; for example, if an author of the new paper was previously mentioned in the group chat history, it will identify the author’s name as a social signal. (3) Team members who might be interested in the paper based on their previous interactions in the chat channel.

2.1.3 Step 3: Generate contextualized message. PAPERPING uses chains of LLM prompts to turn the retrieved social signals into a natural language explanation. PAPERPING uses one prompt for each group of social signals before using one final prompt to succinctly summarize the information and implement a friendly and informative tone. This approach lets the output address meaningful connections between the recommended paper and the channel context while reducing hallucinations.

2.1.4 Step 4: Post to the group and collect continuous feedback. The message gets posted in the group channel, after which group members can provide an emoji reaction or reply to the message to give feedback. Group members can adjust the frequency of recommendations to daily, once every other day, or weekly based on their individual preference.

2.2 PAPERPING System Implementation

PAPERPING is implemented as a Slack application, with recommendations arriving in the group conversation as chat messages from

the PAPERPING bot account. It is built with the Slack Web API. The backend server is built in Node.js and interfaces with a PostgreSQL database. PAPERPING leverages the Semantic Scholar Recommendation API¹ to select paper recommendations based on a group’s interests and prior interactions. To retrieve the paper’s metadata, SPECTER embeddings for each paper [3], and citations from the citation graph, PAPERPING uses the public Semantic Scholar Academic Graph API. Finally, PAPERPING uses GPT4-turbo-preview through the OpenAI API for all text generation with an LLM.

While the current implementation is specific to Slack due to its API design, the underlying architecture of PAPERPING can be extended to other platforms. As long as the system can collect sufficient social signals from group interactions, the same approach of retrieving social signals and generating messages (steps 2 and 3) can be generalized to other collaborative platforms, such as Discord and Microsoft Teams. However, the specific mechanisms for collecting social signals (step 1) and delivering messages (step 4) would need to be reimplemented to align with the APIs and interaction models of each platform.

3 Deployment and Interactive Demo

We have deployed PAPERPING in 18 Slack channels, reaching 500 researchers as part of a user study [14]. As part of the demo, we

¹<https://api.semanticscholar.org/api-docs/recommendations>

would like to invite a broader and more diverse group of users from the CSCW community to explore and use PAPERPING in authentic contexts. We will set up the demo as follows: (1) **Before the conference:** We will advertise the Slack workspace on social media and seed the workspace with example discussions and PAPERPING's interactions. Researchers will be invited to join the workspace and initiate or join the channels that align with their interests. (2) **At the demo desk:** Attendees can browse active and public channels, view live examples of PaperPing's recommendations, and initiate their own discussion topics. Attendees will be encouraged to post, react, and engage with PaperPing and each other in the workspace. The presenter will walk participants through the logic, features, and design choices of PaperPing. This setup will allow participants to experience PaperPing in an authentic context where they can relate to.

Overall, the demo will contribute to the CSCW community by (1) showcasing a functional, socially aware AI agent in action, (2) motivating discussion around the role of AI in group spaces, which is of core interest to CSCW, and (3) providing a platform for researchers to connect with each other and share relevant work on mutual interest, which could potentially continue beyond the event.

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